

Chapter 3: CMOS Transistor Theory

Understanding Transistors via Physical Storytelling & Dynamic Mechanisms

4-Hour Fast-Track Study Guide | Lecture Reference: UIT Chapter 3 & Weste-Harris CMOS VLSI

4-Hour Time-Boxed Roadmap (Lộ trình học tập 4 giờ)

Follow this exact schedule to master Chapter 3 in 4 hours without wasting time on Chapters 1 & 2:

Time Block	Module Name	The Physical Story (Câu chuyện vật lý)
Hour 1 (0h - 1h)	The MOS Capacitor & 3 Modes	The drama of electrostatic forces: How gate voltage drives holes away and creates an electron highway
Hour 2 (1h - 2h)	Derivation of $I - V$ Equations	Traffic flow across silicon: Deriving linear current and the slingshot pinch-off saturation mechanism
Hour 3 (2h - 3h)	pMOS Physics & AMI $0.6\mu\text{m}$ Math	The mobility battle ("Highway" vs "Musical Chairs"): Why pMOS is sluggish and requires double width
Hour 4 (3h - 4h)	Capacitive Parasitics & Speed	The Capacitor Discharge Odyssey: Step-by-step electron loop from Ground to node Y and speed limits

Hour 1: The MOS Capacitor & The 3 Operating Modes

Reference: Slides 1 – 5 | Target Time: 60 minutes

1.1 Beyond the Ideal Switch (Vượt qua tư duy công tắc lý tưởng)

In basic digital logic, a transistor is taught as an ideal switch: Gate = 1 (closed, zero resistance) and Gate = 0 (open, zero current). In real silicon, a transistor is a **charge-controlled reservoir pipe**:

- When turned ON, it does not have zero resistance. Current is limited by carrier collisions in the crystal lattice.
- Every terminal stores electric charge as parasitic capacitance.
- Every switching transition is a race to drain or fill this stored charge:

$$\Delta t = \frac{C}{I} \Delta V$$

1.2 Structure of the MOS Capacitor

The heart of every MOSFET is a parallel-plate sandwich:

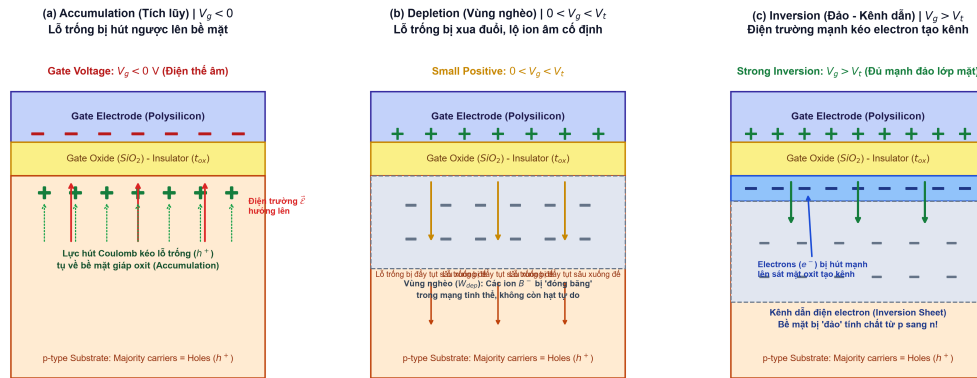
- Gate (Cực Cổng):** Highly conductive Polysilicon (silicon đa tinh thể dẫn điện tốt).
- Dielectric (Lớp điện môi cách điện):** Silicon Dioxide (SiO_2) with thickness t_{ox} and relative permittivity $\epsilon_{ox} = 3.9\epsilon_0$.
- Body / Substrate (Đế bán dẫn):** p-type silicon doped with Acceptor atoms (N_A , e.g., Boron).

The gate oxide capacitance per unit area is:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14} \text{ F/cm}}{t_{ox}}$$

1.3 The Chronological Story of the Three Operating Modes

Imagine you are standing inside the silicon crystal beneath the gate oxide. Here is what happens as gate voltage rises:



Physical Mechanism (Cơ chế vật lý cốt lõi)

1. Act 1: Accumulation (Chế độ tích lũy - $V_g < 0$ V)

- **Electric Field:** Negative voltage on the gate produces an **upward-pointing electric field** ($\uparrow$).
- **Carrier Motion:** Mobile positive holes (h^+) feel an electrostatic pull upward ($(F) = q(\mathcal{E})$). They rush to the surface, forming a dense sheet of holes directly underneath the oxide.
- **Transistor State:** Strictly OFF. There are no free electrons to connect Source to Drain.

2. Act 2: Depletion (Chế độ nghèo - $0 < V_g < V_t$)

- **Electric Field:** Gate voltage turns slightly positive, flipping the field **downward** ($\downarrow$).
- **Carrier Motion:** The positive gate repels positive holes, shoving them deep down into the substrate bulk.
- **Lattice Residue:** As holes retreat, they expose immobile negative Boron acceptor ions (B^-) locked in the crystal lattice. This creates an insulating **Depletion Zone** (W_{dep}) free of mobile carriers.

3. Act 3: Inversion (Chế độ đảo - Hình thành kênh dẫn - $V_g > V_t$)

- **Electric Field:** When V_g exceeds the **Threshold Voltage** (V_t), the downward electric field becomes overwhelming ($> 10^6$ V/cm).
- **Carrier Motion:** The field forcibly pulls minority electrons from the substrate and source/drain diffusions, locking them into a razor-thin 2D sheet ($\approx 1 - 3$ nm) at the surface.
- **The Magic:** The surface silicon has inverted from p-type to n-type! A conducting bridge of electrons is born.

VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

Physical Meaning of Threshold Voltage V_t :

V_t is not an arbitrary number. It is the exact gate voltage needed to bend the energy bands enough so that the surface electron concentration equals the bulk hole concentration ($n_{\text{surface}} = N_A$). Below V_t , the channel does not exist!

Hour 2: Mathematical Derivation of Shockley $I - V$ Equations

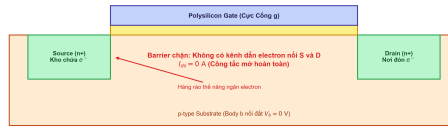
Reference: Slides 6 – 14 | Target Time: 60 minutes

2.1 The Battle of Two Electric Fields

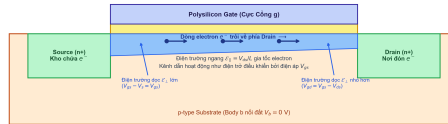
When a channel forms, current flow is governed by the tug-of-war between two perpendicular electric fields:

1. **Vertical Field ($\mathcal{E}_\perp$):** Created by Gate-to-Channel voltage. Controls **how many electrons** are in the channel.
2. **Lateral Field ($\mathcal{E}_\parallel$):** Created by Drain-to-Source voltage ($\mathcal{E}_\parallel = \frac{V_{ds}}{L}$). Propels electrons **from Source to Drain**.

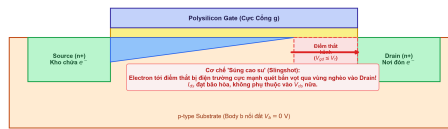
1. Cutoff ($V_{gs} < V_t$): Kênh chưa mở | Không có electron tại bề mặt | $I_{ds} = 0$



2. Linear ($V_{gs} > V_t$, $V_{ds} < V_{gs} - V_t$): Kênh liên tục | Dòng electron chạy đều từ S sang D



3. Saturation & Pinch-off ($V_{gs} > V_t$, $V_{ds} \geq V_{gs} - V_t$): Thắt kênh | Cơ chế 'Súng cao su' quét electron



2.2 First-Principles Derivation of Current (I_{ds})

Current is charge passing through a cross-section per unit time: $I_{ds} = \frac{Q_{\text{channel}}}{t_{\text{transit}}}$.

Mathematical Derivation (Chứng minh toán học)

Step 1: Calculate Total Channel Charge (Q_{channel})

At the Source ($x = 0$), channel voltage is $V_s = 0$. Gate-to-channel voltage is V_{gs} .

At the Drain ($x = L$), channel voltage is $V_d = V_{ds}$. Gate-to-channel voltage is $V_{gd} = V_{gs} - V_{ds}$.

The channel tapers into a wedge! The average channel voltage along the length is $V_{\text{channel, avg}} = \frac{V_s + V_d}{2} = \frac{V_{ds}}{2}$.

Effective inversion voltage: $V_{\text{eff}} = \left(V_{gs} - \frac{V_{ds}}{2}\right) - V_t$.

$$Q_{\text{channel}} = C_{ox} W L \left[(V_{gs} - V_t) - \frac{V_{ds}}{2} \right]$$

Step 2: Calculate Carrier Transit Time (t_{transit})

Lateral electric field is $\mathcal{E}_\parallel = \frac{V_{ds}}{L}$. Electron drift velocity is $v = \mu_n \mathcal{E}_\parallel = \mu_n \frac{V_{ds}}{L}$.

Transit time across distance L : $t_{\text{transit}} = \frac{L}{v} = \frac{L^2}{\mu_n V_{ds}}$.

Step 3: Combine to Obtain Linear Current

$$I_{ds} = \frac{Q_{\text{channel}}}{t_{\text{transit}}} = \mu_n C_{ox} \frac{W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

2.3 The Slingshot Mechanism at Pinch-off (Cơ chế bắn vọt khi thắt kênh)

When V_{ds} increases such that $V_{gd} = V_{gs} - V_{ds} \leq V_t$, the vertical field at the drain falls below the threshold needed for inversion. The inversion layer disappears at the drain edge. This is **Pinch-off** (thắt kênh).

Physical Mechanism (Cơ chế vật lý cốt lõi)

Why Current Does NOT Stop at Pinch-off:

Electrons entering from the Source travel through the channel wedge. Upon reaching the pinch-off point, they encounter a fierce, localized electric field across the depletion gap. This field acts like a **slingshot (súng cao su)**, violently sweeping electrons across into the n^+ Drain!

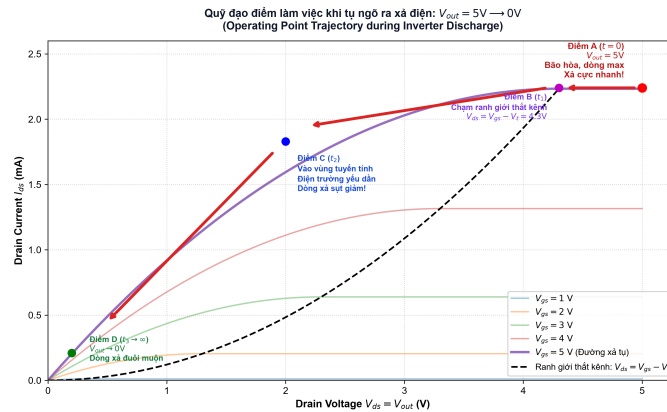
Increasing V_{ds} further simply widens this tiny depletion gap slightly. The voltage drop across the active channel remains locked at $V_{dsat} = V_{gs} - V_t$. Therefore, **current completely saturates!**

Substituting $V_{ds} = V_{gs} - V_t$ into the linear equation gives the constant saturation current:

$$I_{ds, \text{sat}} = \frac{\beta}{2} (V_{gs} - V_t)^2 \quad \text{where} \quad \beta = \mu_n C_{ox} \frac{W}{L}$$

2.4 Dynamic Operating Trajectory During Inverter Switching

Notice how an actual circuit moves on the $I - V$ plane when an inverter discharges an output load:



⚡ VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

Reading the Trajectory from Right to Left:

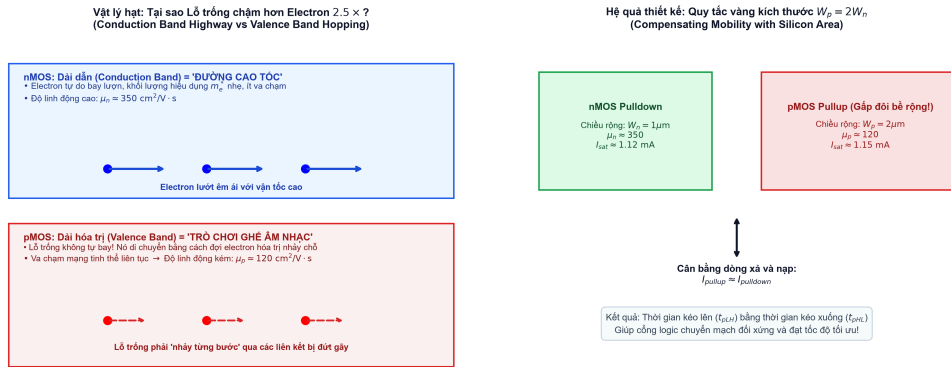
- **Point A ($t = 0$):** $V_{out} = V_{ds} = 5V$. Transistor is in **Saturation**. It delivers maximum current ($\approx 2.24mA$), discharging the load capacitor at maximum speed ($d\frac{V}{dt} = -\frac{I_{sat}}{C}$).
- **Point B ($t = t_1$):** Output drops to $V_{dsat} = 4.3V$. Transistor crosses the pinch-off boundary.
- **Points C $\rightarrow$ D ($t > t_1$):** Transistor enters the **Linear Region**. As $V_{out} \rightarrow 0$, the driving electric field collapses, current decays, and the discharge slows down to an exponential tail.

Hour 3: pMOS Physics & AMI 0.6 μm Hand Calculation

Reference: Slides 15 – 17 | Target Time: 60 minutes

3.1 The Microscopic Mobility Battle (“Highway” vs “Musical Chairs”)

In silicon, an nMOS transistor conducts via **electrons**, while a pMOS transistor conducts via **holes**:



Physical Mechanism (Cơ chế vật lý cốt lõi)

Why Are Holes 2.5x Slower Than Electrons?

- **Electrons (The Open Highway):** Free electrons travel in the **Conduction Band** (dải dẫn). They glide through empty energy space with light effective mass ($m_e^* \approx 0.26m_0$) and experience low lattice collisions. Mobility is high: $\mu_n \approx 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$.
- **Holes (The Game of Musical Chairs):** Holes do not physically exist as free particles. A hole is an **empty covalent bond** in the **Valence Band** (dải hóa trị). For a hole to move, a bound electron must break its bond and hop into the empty chair. This constant lattice scattering creates high resistance: $\mu_p \approx 120 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$.

VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

The Inverter Sizing Golden Rule ($\frac{W_p}{W_n} = 2$):

Because holes travel roughly half as fast as electrons ($\frac{\mu_n}{\mu_p} \approx 2 - 3$), a pMOS transistor must be designed with **twice the channel width** ($W_p \approx 2W_n$) to deliver equal drive current. This ensures equal pull-up and pull-down propagation delays ($t_{pLH} \approx t_{pHL}$).

3.2 Worked Hand Calculation: AMI 0.6 μm Process (Directly solving Slide 15 from the university lecture)

Given Parameters:

- Process Node: 0.6 μm , Gate oxide: $t_{ox} = 100 \text{ \AA} = 100 \times 10^{-8} \text{ cm} = 10 \text{ nm}$.
- Electron mobility: $\mu_n = 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$.
- Threshold voltage: $V_t = 0.7 \text{ V}$. Transistor geometry: $\frac{W}{L} = \frac{4\lambda}{2\lambda} = 2$.

Mathematical Derivation (Chứng minh toán học)

Step 1: Calculate Oxide Capacitance per Area (C_{ox})

$$\varepsilon_{ox} = 3.9 \times \varepsilon_0 = 3.9 \times (8.85 \times 10^{-14} \text{ F/cm}) = 3.45 \times 10^{-13} \text{ F/cm}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-13} \text{ F/cm}}{100 \times 10^{-8} \text{ cm}} = 3.45 \times 10^{-7} \text{ F/cm}^2$$

Step 2: Calculate Process Transconductance Parameter ($k'_n = \mu_n C_{ox}$)

$$k'_n = \mu_n C_{ox} = 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \times (3.45 \times 10^{-7} \text{ F/cm}^2) = 120.8 \mu\text{A/V}^2$$

Step 3: Calculate Transistor Gain Factor (β)

$$\beta = k'_n \frac{W}{L} = 120.8 \mu\text{A/V}^2 \times 2 = 241.6 \mu\text{A/V}^2 = 0.242 \text{ mA/V}^2$$

Step 4: Calculate Saturation Current at $V_{gs} = 5 \text{ V}$

$$V_{dsat} = V_{gs} - V_t = 5.0 \text{ V} - 0.7 \text{ V} = 4.3 \text{ V}$$

$$I_{ds, \text{sat}} = \frac{\beta}{2} (V_{gs} - V_t)^2 = \frac{0.242 \text{ mA/V}^2}{2} \times (4.3 \text{ V})^2 = 0.121 \times 18.49 = 2.24 \text{ mA}$$

Comparison: pMOS Equivalent for AMI 0.6 μm

For a pMOS in the same process ($\mu_p = 120 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$, $V_{tp} = -0.7 \text{ V}$, $\frac{W}{L} = 2$):

$$\beta_p = \mu_p C_{ox} \frac{W}{L} = 120 \times (3.45 \times 10^{-7}) \times 2 \approx 0.083 \text{ mA/V}^2$$

At $V_{gs} = -5 \text{ V}$:

$$I_{ds, \text{sat}} = -\frac{\beta_p}{2} (|V_{gs}| - |V_{tp}|)^2 = -\frac{0.083}{2} \times (4.3)^2 \approx -0.77 \text{ mA}$$

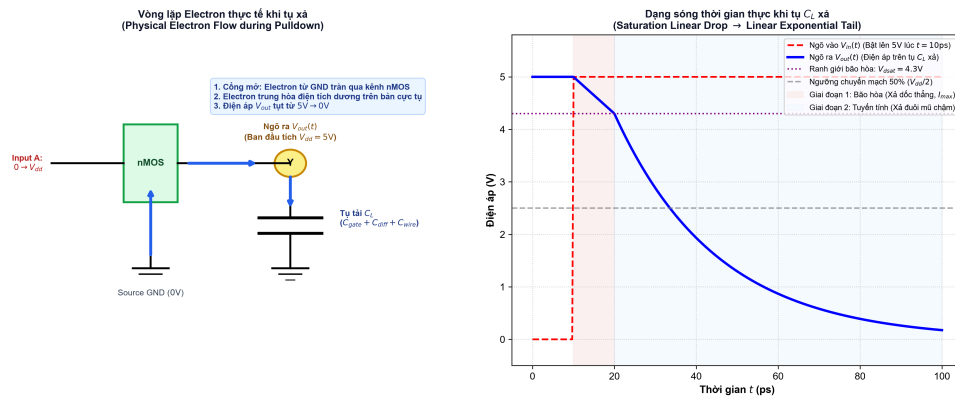
Notice that $\frac{I_{ds, n}}{|I_{ds, p}|} = \frac{2.24}{0.77} \approx 2.9 \times$, mathematically proving why pMOS must be sized wider!

Hour 4: Transistor Parasitic Capacitance & Circuit Speed

Reference: Slides 18 – 20 | Target Time: 60 minutes

4.1 The Capacitor Discharge Odyssey (Cuộc hành trình xả tụ ở cấp độ nguyên tử)

What actually happens when an inverter output node switches from 1 $\rightarrow$ 0?



Physical Mechanism (Cơ chế vật lý cốt lõi)

The 4 Acts of Node Discharge:

- Act 1 (The Stored Charge):** The load capacitor C_L (sum of gate, diffusion, and wire capacitances) stores positive charge on node Y ($Q = C_L V_{dd}$).
- Act 2 (Gate Inversion):** Input V_{in} steps from 0 $\rightarrow$ V_{dd} . In picoseconds, a conductive electron channel bridges Source and Drain.
- Act 3 (The Electron Surge from Ground):** Ground is an infinite reservoir of electrons. The positive potential at node Y creates an electric field that **drags electrons from Ground, up through the nMOS channel, onto the capacitor plate** to neutralize the charge.
- Act 4 (Lattice Collisions & Speed Limit):** Why doesn't it discharge in 0 seconds? As electrons sprint through silicon, they collide with vibrating silicon atoms (phonons). These collisions create **Channel Resistance (R_{on})**, governing the delay: $\Delta t = \frac{C_L \Delta V}{I_{avg}}$.

4.2 Gate vs. Parasitic Diffusion Capacitance

- Gate Capacitance ($C_g = C_{ox} WL$):** Functional and desirable. It forms the channel and presents $\approx 1.5 - 2.0$ fF/ μm load to the driver.
- Diffusion Capacitance (C_{sb}, C_{db}):** Purely parasitic. Caused by reverse-biased PN junction diodes between n^+ diffusions and the p-body. **Layout tip:** Share diffusion contacts to cut this parasitic capacitance by 50%.

4.3 Summary Cheat Sheet (Bảng tóm tắt toàn chương)

Symbol / Parameter	Physical Meaning	Typical Value / Formula
C_{ox}	Gate oxide capacitance per area	$\frac{\epsilon_{ox}}{t_{ox}} \approx 3.45 \times 10^{-7} \text{ F/cm}^2$
β	Transconductance gain factor	$\mu C_{ox} \left(\frac{W}{L} \right)$
Linear Region	Continuous resistive channel	$V_{gs} \geq V_t$ and $V_{ds} < V_{gs} - V_t$
Saturation Region	Channel pinched-off at drain	$V_{gs} \geq V_t$ and $V_{ds} \geq V_{gs} - V_t$
$I_{ds, \text{sat}}$	Shockley saturation current	$\frac{\beta}{2} (V_{gs} - V_t)^2$
$\frac{\mu_n}{\mu_p}$	Mobility ratio (electrons vs holes)	$\approx 2 \text{ to } 3 \rightarrow W_p = 2W_n$
Δt	Gate propagation delay	$\frac{C_L \Delta V}{I_{\text{avg}}}$

4.4 10-Minute Self-Assessment Quiz (Bài kiểm tra nhanh)

- Q1:** An nMOS transistor has $V_{gs} = 3.0 \text{ V}$, $V_t = 0.6 \text{ V}$, and $V_{ds} = 1.5 \text{ V}$. Which operating region is it in?
Answer: $V_{\text{dsat}} = V_{gs} - V_t = 3.0 - 0.6 = 2.4 \text{ V}$. Since $V_{ds} = 1.5 \text{ V} < 2.4 \text{ V}$, the channel is continuous and the transistor is in the **Linear Region**.
- Q2:** Why does drain current remain constant in the saturation region even if V_{ds} increases?
Answer: The channel pinches off at the drain side. The voltage across the active inversion channel is clamped at $V_{\text{dsat}} = V_{gs} - V_t$. Extra voltage drops across the pinch-off depletion gap, acting as a slingshot without increasing channel field.
- Q3:** Why is pMOS designed twice as wide as nMOS ($W_p \approx 2W_n$) in standard CMOS cells?
Answer: Hole mobility in silicon is $2 - 3 \times$ lower than electron mobility ($\frac{\mu_n}{\mu_p} \approx 2 - 3$) due to bond-hopping collisions in the valence band. A $2 \times$ wider pMOS matches nMOS drive current for symmetric delays ($t_{pLH} \approx t_{pHL}$).
- Q4:** During capacitor discharge, why does discharge speed slow down near the end?
Answer: In the beginning (Saturation), current is constant at I_{sat} (V_{out} drops linearly). Once $V_{\text{out}} < V_{gs} - V_t$, the transistor enters the Linear region. Here, current is proportional to the remaining voltage, decaying exponentially toward zero.